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School of Computing
First CIA Exam –Sep.-2023
Course Code: -INT434
Course Name: Adv.Social Text and Media Analytics
Duration: 90 minutes
Max Marks: 50

PART – A 5 x 10 =50

Answer all the questions

1. Explain with a detailed note on purpose of classification and also elaborate KNN classification algorithm and perform the same on the following data set and predict the class for the following
   d. Maths=6, computers=8
   e. Maths=4 , computers =7
   f. Maths =7 , computers=3

2. Making use of the given data set apply Naive Bayes algorithm and predict that if a fruit has the following property then which type of fruit it is
   a.Fruit = { Yellow,Sweet,Long}
   b. Fruit ={Yellow ,Sweet. Short}

Fruit	Yellow	Sweet	Long	total
Mango	35	45	0	65
Banana	40	30	35	40
Orange	5	10	5	15
total	80	85	40	120

3. Illustrate the role and importance of Apriori algorithm and apply the same to measure the associations between data items
   Min. Support=45%
   Threshold confidence = 70%

Transactions	Items
1	A ,C,D
2	B,C,E
3	A,B,C,E
4	C,D,E
5	B,E

4. Explain NLP in detail the 1) Componenets 2) levels/phases 3)IR with a suitable real time example.
5. Apply Agglomerative Hierarchical clustering after explaining the same for the following set of one dimensional data points
   18,20,22,25,34,27,42,44,38,41
   Also draw the corresponding dendogram.

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School of Computing
First CIA Exam – Feb 2025
Course Code: INT434
Course Name:Advanced Social,Text and Media Analytics
Duration: 90 minutes Max Marks: 50

PART A

Answer ALL the questions **5x10=50 marks.**

1. a. Summarize the basic principles of Vector Space model in information retrieval with suitable example.

b. Using the text mining method of TF-IDF, calculate the TF and IDF for the following small set of documents:Doc 1: "Text mining is fun".Doc 2: "Mining data with text is important".Doc 3: "Text data and mining techniques" (Show all calculations for TF and IDF and compute the final TF-IDF scores for the word "text" in each document.)

2. Calculate entropy and purity for the given group of clusters.

Cluster Id:	A	B	C
TOPIC 1:	250	20	10
TOPIC 2:	20	180	80
TOPIC 3 :	30	100	210

3. Using the IQR method, describe the steps involved in detecting outliers in a given dataset. Illustrate your explanation with the following dataset:4,8,10,14,16,18,20,22,24,28

i.Calculation of Q1, Q3, and IQR. ii.Determining lower and upper bounds for outlier detection. iii.Identify any outliers in the dataset.

4. Write in detail condition random field probabilistic model for information extraction .

5. Explain the following with suitable examples

a. Shallow Parsing. b. Constituency Grammars c. Dependency Grammar

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School of Computing
Second CIA Exam –Oct.-2023
Course Code:- INT434
Course Name: Adv.Social Text and Media Analytics
Duration: 90 minutes
Max Marks: 50

PART – A 4 x 10 =40

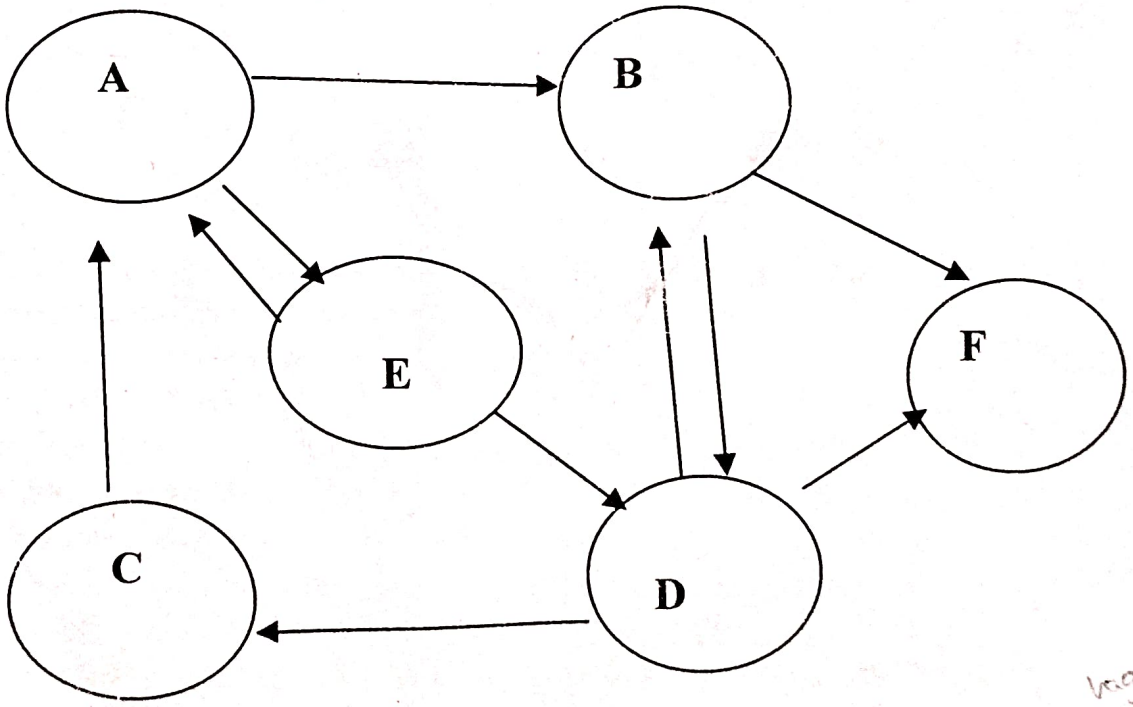
Answer any four questions

1. Explain with a detailed note with suitable example sentiment analysis
2. Explain Topic detection and also explain with a suitable example LDA Algorithm
3. Identify the need , importance and role of Clickstream analysis in detail
4. Interpret the meaning and need of SEO in detail with a suitable model.
5. Explain with suitable example the terms web crawling, indexing, ranking. Also explain on web traffic models and their significance.

PART – B 1 x 10 =10

Answer the following question

6. Solving the dead end problem in the following graph find the page rank for each page using matrix method. Also compare the result with numerical formula method.(3 iterations)



80C 415 / ASTMA -2

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School of Computing
Second CIA Exam – March 2025
Course Code: INT434
Course Name: Advanced Social Text Media Analytics
Duration: 90 minutes Max Marks: 50

PART A

Answer any TWO questions **2x15=30 marks**

1. Illustrate the **'big three centrality measures'** used in social network analysis with examples.

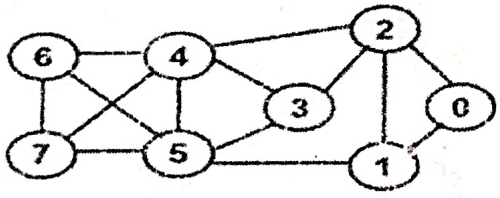
2. a. Given the following undirected graph: Vertices: A, B, C, D, E, F, Edges: AB, BC, BE, BD, CD, CE, DE, DF. Compute the k-core collapse sequence for k=1,2,3, and 4. **(8marks)**

b. Describe the importance of Off-Page SEO and how they influence search engine rankings. **(7marks)**

3. Write short notes on :

a. Two-mode network analysis with examples **(6marks)**

b. Find the Egocentric network of all nodes. **(9 marks)**



Part – B **1x20=20 marks**

4. A social media analysis categorizes user interactions into two sentiment states: **Engaged (E)** and **Disengaged (D)**. The initial probabilities of user sentiment states are P(E)=0.6 and P(D)=0.4. Over time, user sentiment transitions according to the following probabilities: P(E→E)=0.7, P(E→D)=0.3, P(D→D)=0.6, and P(D→E)=0.4. Users express their sentiments through their **Activity Type**: either **Comments (C)** or **Shares (S)**. The likelihood of these activity types, given the sentiment state, is described by the emission probabilities P(C|E)=0.8, P(S|E)=0.2, P(C|D)=0.4, and P(S|D)=0.6. Given the observed sequence of activities [C,S,S], use the Hidden Markov Model (HMM) to identify the most likely sequence of user sentiment states. Calculate the probability of the observation sequence using the Forward Algorithm.

SASTRA DEEMED TO BE UNIVERSITY THINK SUBTLE | THINK TRANSPARENCY | THINK SASTRA	School of Management First CIA Test – February 2024 Yr. Prog: Semester VIII -B.Tech CSBS Course Code: MGT214 Course Name: Marketing Research and Marketing Management Duration: 90 minutes Max Marks: 50

PART A

Answer all the questions **10x2=20 marks**

1.	Relate the significance of the 7Ps of Marketing.
2.	List out any three core concepts in marketing.
3.	What does a holistic marketing philosophy include?
4.	How does marketing affectcustomer value?
5.	Why should we understand the behavior of consumers in the market?
6.	Explain New Product Development.
7.	Identify the characteristics of Services.
8.	Explain the role of packaging in Marketing.
9.	'The customer is the King'. Justify with relevant examples.
10.	What external factors affect the working of an organization?

PART B

Answer all the questions **3 x 10 = 30 Marks**

11.	Consumer behavior is important in understanding the buying decisions of the consumer. Analyze the various factors that influence the consumer.
12.	Elucidate Market segmentation and its types with suitable examples.
13.	Explain the various stages of the Product Life cycle with relevant examples.

SOC - 319

MRMM - 08.

SASTRA DEEMED TO BE UNIVERSITY	School of Management Second CIA Test – March 2024 Yr. Prog: Semester VIII -B.Tech CSBS Course Code: MGT214 Course Name: Marketing Research and Marketing Management Duration: 90 minutes Max Marks: 50

PART A

Answer all the questions **10x2=20 marks**

1.	Relate the significance of retailing
2.	List out the objectives of the advertising strategy.
3.	How can businesses utilize influencers for publicity purposes?
4.	List out the methods of data collection for market research.
5.	How does e-commerce impact traditional brick-and-mortar retail?
6.	Explain the difference between push and pull marketing strategies.
7.	What is the purpose of marketing research in business?
8.	Differentiate factor and cluster analysis with examples.
9.	What is the purpose of a questionnaire in marketing research?
10.	Bring out the significance of Marketing Communications.

PART B

Answer all the questions **3 x 10 = 30 Marks**

11.	Explain the various steps involved in the marketing research process.
12.	Examine the various types of channels and the factors that must be considered while making Channel Selection.
13.	Describe what exploratory, descriptive, and causal research is and how they are related to one another. Provide an example of each type of research.

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School of Computing
Third CIA Exam –May 2025
Course Code: MGT214
Course Name: Marketing Research & Marketing Management
Duration: 90 minutes Max Marks: 50

PART A

Answer any 5 questions **5x10=50 Marks**

1. Explain the different functions of marketing and describe how changing business environments are influencing modern marketing practices.
2. Explain the stages of the new product development process and describe what strategies should be emphasized at each stage.
3. Explain the different types of market research process
4. In what ways has digital marketing revolutionized traditional marketing practices? Discuss with relevant examples to illustrate its impact.
5. Discuss key design elements that contribute to creating effective and user-friendly websites with examples?
6. Explain the 7 Ps of marketing and illustrate each element with suitable examples
7. Enumerate the different components of internet marketing with examples

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School of Computing
First CIA Exam – Feb 2025
Course Code: MGT213
Course Name: Service Science and Service Operations Management
Duration: 90 minutes Max Marks: 50

PART A

Answer the following questions **5x10=50**

1.Discuss the role of service in economy and society.

2.Elucidate the nature of services and service encounters.

3.Explain various frameworks to design service operation systems.

4. Evaluate Goods dominant logic to Service dominant logic.

5.Elaborate the strategies for matching capacity and demand.

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School of Computing
Second CIA Exam – Mar 2025
Course Code: MGT213
Course Name: Service Science and Service Operations Management
Duration: 90 minutes Max Marks: 50

PART A

Answer the following questions **5x10=50**

1. Discuss the NSD Cycle.
2. Elaborate Complaint handling & Service recovery.
3. Evaluate forecasting demand & service.
4. Explain Vehicle routing problem.
5. Elucidate managing Service supply relationship.

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School of Computing
First CIA Exam – February 2024
Course Code: INT422
Course Name: IT Project Management
Duration: 90 minutes Max Marks: 50

LTC204

ITPM-6

PART A

Answer all the questions **5 * 10 = 50 marks**

1. Explain in detail about project overview and feasibility studies.
2. What is project management? List all the key aspects of the software project management with their purpose.
3. How to identify a software project and how to formulate a software project leading to a product launch? Explain both with example of Company ABC trying to launch android billing application for supermarket.
4. Explain in detail about Software project cost estimation with an example for the same project given in question 3.
5. Distinguish the difference between PERT and CPM. List the applications of both. Discuss the role of network diagram in PERT and CPM. How to draw the network diagram?

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School of Computing
Second CIA Exam – March 2025
Course Code: INT422
Course Name: IT Project Management
Duration: 90 minutes Max Marks: 50

PART A

Answer ANY FIVE questions **5 x 10 = 50 Marks**

1. Describe Project Control within the context of IT project management. Discuss its essential elements, identify relevant tools and techniques, outline key activities involved, and analyze the benefits gained from implementing effective project control.
2. Explain how resources are allocated in software product development. Describe the processes of resource scheduling and resource leveling within the context of an IT project, and identify the key elements involved in each process.
3. Describe the concept of Risk in Software Project Management (SPM). Explain how risks can be managed effectively during software product development, and outline the steps involved in the risk management process within SPM.
4. Explain the concept of a project audit in software product development and justify its necessity. Describe the detailed

procedure for conducting a project audit, and outline the processes involved in both project audit and project termination.

5. Describe the Agile approach in software development. Differentiate Agile from traditional software development models. Identify various types of Agile methodologies and explain any six principles underpinning the Agile framework.
6. Explain the Scrum framework within software development. Identify and describe the primary roles involved, and outline the essential terminologies associated with Scrum methodology.

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School of Computing
Second CIA Exam – February 2024
Course Code: INT422
Course Name: IT Project Management
Duration: 90 minutes Max Marks: 50

PART A

Answer all the questions **5 * 10 = 50 marks**

1. You have a list of tasks (activities) to find the Critical path for the project XYZ. Find the critical, noncritical activities, float of each activity, and critical path using AON Diagram.

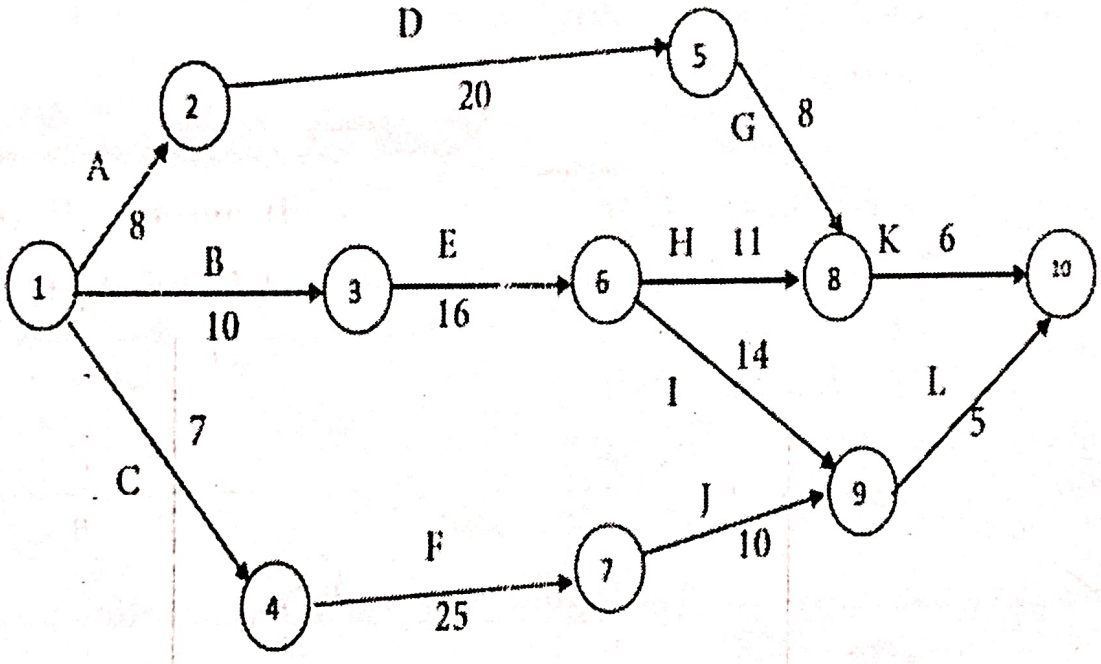
Task ID	Task Description	Task Predecessors	Task Duration (hours)
A	Project start		0
B	Buy materials for A	A	10
C	Buy materials for B	A	20
D	Build A	B, C	30
E	Build B	B, C	20
F	Polish and finish B	E	40
G	Join A and B	D, F	20
H	Project finish	G	0

2. Explain the Resource Scheduling and Resource Leveling procedure for software project management.
3. How to manage risk in SPM? Explain the process of analyzing risks and project control.
4. How to do the project audit? When a software project should be terminated? Explain the possibilities for project termination.

5. A) Determine the critical path, the critical activities and the project completion time.

Activity	Predecessor Activity	Duration (Weeks)
A	-	3
B	A	5
C	A	7
D	B	10
E	C	5
F	D,E	4

B) Find out the completion time and the critical activities for the following project:



ITC-412
ITPM-8.

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School of Computing
Third CIA Exam – April 2024
Course Code: INT422
Course Name: IT Project Management
Duration: 90 minutes Max Marks: 50

PART A

Answer all the questions **5 * 10 = 50 marks**

1. How to identify the Market for the software development? How to perform Demand analysis for the same.
2. Explain in detail about Resource scheduling and leveling. Justify the need for both SPM.
3. List and explain various terminologies used in Scrum with neat sketch.
4. Narrate the functional behavior of DevOps and its components along with Version Control.
5. Write short notes on:
   - Continuous Integration
   - Continuous Delivery
   - Continuous Deployment
   - Automated Monitoring.

compulsory

5) write short notes on all the other agile methodologies.

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School of Computing
Third CIA Exam – April 2025
Course Code: INT422
Course Name: IT Project Management
Duration: 90 minutes Max Marks: 50

PART A

Answer ANY FIVE questions **5 x 10 = 50 Marks**

1. List all the key components of project overview. Explain each of them with their purposes.
2. Develop a Project Schedule for the Development of a Web Application for the Online Fruits selling Business. Create a schedule for every phase of the project along with the respective tasks to completed along with task dependencies. Use any of the visualization method for the schedule.
3. Explain all the key components of project control with their purposes along with benefits of the project control activities in Software project Development.
4. Build a Web Application as per the Project Schedule created in Question number 2 using Scrum. Plan the development and explain each sprint as per the plan.